Feature Selection in Statistical Machine Learning

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Lecture 3

Outline

- Mathematical formulation re-visited and recap
- Conditional independence perspective
- Diving deeper into conditional independence

Problem Statement

Data: (p features and n samples)

$$(\{X_j^{(i)}\}_{j=1}^{p}, Y^{(i)})_{i=1}^{n}$$

E.g. Housing data: $n = 300$, $p = 3$
Informal goal: Given the set of p features, find the s features that directly contribute to the prediction of Y.

Example 1: Housing data

$n = 300$
$p = 3$ (square footage, number of bedrooms, neighborhood quality)
$s = 2$ (square footage, neighborhood quality)

Example 2: Protein function response

$n \approx 2 \times 10^6$ (number of protein sequences)
$p \approx 500^{20}$ (different permutations of single sequence)
$s \approx 10$ (set by practitioner)

Example 3: MNIST image data

$n \approx 50,000$ (each image is a sample)
$p = 784$ (28×28) (assuming each image pixel value is a feature)
$s \approx 20 - 100$ (depending on how much loss we allow)

Mathematical formulation: Loss function/prediction perspective

In the *population* or *ideal* setting where there is infinite data:

$$\mathbb{E}[Y|X_1, X_2, ..., X_p] = f(X_1, X_2, ..., X_p)$$

$$S^* = \arg \min_{S \subseteq \{1,2,...,p\}, |S| \leq s} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

Population and finite sample

General feature selection:

$$S^* \in \arg \min_{S \subseteq \{1,2,\ldots,p\}, |S| \leq k} \min_{f \in \mathcal{F}} \mathbb{E}[\mathcal{L}(Y, f(X_S))]$$

Feature selection for this example:

$$S^* \in \arg \min_{S \subseteq \{X_1, X_2, X_3\}, |S| \leq s} \min_{\beta_S} \mathbb{E}[(Y - \sum_{j \in S} X_j \beta_j)^2]$$

Empirical (estimated quantities):

$$\widehat{S^*} \in \arg \min_{S \subseteq \{1,2,\ldots,p\}, |S| \leq k} \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(Y^{(i)}, f(X_S^{(i)}))$$

Housing data example:

$$\widehat{S^*} \in \arg \min_{S \subseteq \{1,2,\ldots,p\}, |S| \leq s} \min_{\beta_S} \frac{1}{n} \sum_{i=1}^{n} (Y^{(i)} - \sum_{j \in S} X_j^{(i)} \beta_j)^2$$

Take-aways from formulation

- Need to simultaneously perform prediction and feature selection
- Two central questions/challenges:
  - **Computational:** Can we compute $\widehat{S^*}$ in reasonable time?
  - **Statistical:** How close is $\widehat{S^*}$ to S^*?

Example: Housing data

$n = 300$

$p = 3$ (square footage, number of bedrooms, neighborhood quality)

$S^* = (X_1, X_3)$

For correlation, $\widehat{S^*} = (X_1, X_2, X_3)$

For regression, $\widehat{S^*} = (X_1, X_3)$

Issues with this formulation

- What if we don't know s?
- How do we efficiently check all $\binom{p}{s}$ subsets when p is large?
- How do we account for noise/sampling error, e.t.c?

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Alternative view: Conditional independence perspective

$$S^* \in \arg \min_{S \subseteq \{1,2,...,p\}, Y \perp X_{S^c} | X_S} |S|$$

were $S^c = \{1, 2, ..., p\}/S$ and $|S|$ refers to the size of S.

- What does the symbol $\perp$ and $|$ mean?
- Note that no explicit assumptions required on prediction function (although it is hidden) in these symbols
- Model-agnostic
- We will slowly work through this definition

Conditional independence in words

$$Y \perp X_A | X_B$$

Y does not depend on the variables X_A given knowledge of the X_B variables

Including the X_A variables does not improve prediction of Y given X_B is already in the model

Opposite is *conditional dependence:*

$$Y \not\perp X_A | X_B$$

Y depends on the variables X_A given knowledge of the X_B variables

Including the X_A variables improves prediction of Y given X_B is already in the model

Example: Housing data

Y : House price; X_1 : Square footage; X_2 : Number of bedrooms; X_3 : Neighborhood quality score

$$S^* = \arg \min_{S \subseteq \{1,2,\ldots,p\}, Y \perp X_{S^c} | X_S} |S|$$

$$S^* = (1, 3)$$

i.e.

$$Y \perp X_2 | (X_1, X_3)$$

In words: *House price does not depend on number of bedrooms given that we know the square footage and neighborhood quality score*

Equivalence between conditional independence and loss function perspectives

$$Y \perp X_{S^c} | X_S \Leftrightarrow \mathbb{E}[Y|X_1, X_2, ..., X_p] = \mathbb{E}[Y|X_S] = f(X_S)$$

Finding minimal subset for given function and loss function is equivalent to finding minimial subset that gives conditional independence.

Now let's take a step back

Independence, dependence of events in probability.
Two events A and B are independent if

$$P(A \cap B) = P(A).P(B)$$

Example: Consider 2 fair 2-sided coins and define 3 random variables:

$$\begin{aligned} A &: \text{First coin is head} \\ B &: \text{Second coin is head} \\ C &: \text{Both coins are the same} \end{aligned}$$

Note that $P(A) = P(B) = P(C) = \frac{1}{2}$

Independence of events example

For events A, B and C we can show that all pairs A, B and C are independent.

$$P(A \cap B) = P(A).P(B)$$
$$P(A \cap C) = P(A).P(C)$$
$$P(B \cap C) = P(B).P(C)$$

Conditional probability and conditional independence

For events A, B and C we showed that all pairs of events are independent.
Now we move on to conditional independence.
Two events A and B are conditionally independent given C if

$$P(A \cap B|C) = P(A|C).P(B|C)$$

Computing $P(A|B)$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Conditional probability example

Example: Consider 2 fair 2-sided coins and define 3 random variables:

$$
\begin{aligned}
A &: \text{ First coin is head} \\
B &: \text{ Second coin is head} \\
C &: \text{ Both coins are the same}
\end{aligned}
$$

$$
\begin{aligned}
P(A \cap B|C) &= P(A|C).P(B|C) \\
P(A \cap C|B) &\neq P(A|B).P(C|B) \\
P(B \cap C|A) &\neq P(B|A).P(C|A)
\end{aligned}
$$

Random variable extension

Example: Consider 2 fair 2-sided coins and define 3 random variables:

$$
\begin{aligned}
X_A = 1 &\quad : \quad \text{First coin is head} \\
X_B = 1 &\quad : \quad \text{Second coin is head} \\
X_C = 1 &\quad : \quad \text{Both coins are the same}
\end{aligned}
$$

and if these events do not occur, we assume these random variables are 0. It follows that for *any* $x_A, x_B, x_C \in \{0, 1\}$:

$$
\begin{aligned}
P(X_A = x_A \cap X_B = x_B | X_C = x_C) &= P(X_A = x_A | X_C = x_C).P(X_B = x_B | X_C = x_C) \\
P(X_A = x_A \cap X_C = x_C | X_B = x_B) &\neq P(X_A = x_A | X_B = x_B).P(X_C = x_C | X_B = x_B) \\
P(X_B = x_B \cap X_C = x_C | X_A = x_A) &\neq P(X_B = x_B | X_A = x_A).P(X_C = x_C | X_A = x_A)
\end{aligned}
$$

Note: Conditional independence needs to hold for all subsets of values when discussing random variables/features.

Independence and conditional independence for random variables

Let X_1 and X_2 be two random variables. X_1 and X_2 are independent if:

$$P(X_1 \in A_1 \cap X_2 \in A_2) = P(X_1 \in A_1).P(X_2 \in A_2)$$

for *every* subset A_1, A_2 in $\mathbb{R}$.
Let X_3 be an additional random variable. X_1 and X_2 are independent conditional on X_3 if:

$$P(X_1 \in A_1 \cap X_2 \in A_2 | X_3 \in A_3) = P(X_1 \in A_1 | X_3 \in A_3).P(X_2 \in A_2 | X_3 \in A_3)$$

for *every* subset A_1, A_2, A_3 in $\mathbb{R}$.

Independence and conditional independence in linear model(s)

Consider the following linear model (X_1, X_2, X_3, Y):

$$\begin{aligned} Y &= X_1 + X_3 + \epsilon_Y \\ X_2 &= \frac{1}{2}X_1 + \epsilon_2 \end{aligned}$$

$(X_1, \epsilon_2, X_3, \epsilon_Y)$ are all *independent* zero-mean random variables.
Question: Which features (X_1, X_2, X_3) is Y dependent on?
Question: Which features (X_1, X_2, X_3) is Y *conditionally* dependent on given all the other features?